

SB Stocks - Project Documentation

Date	Team ID	Project Name	Maximum Marks
		SB Stocks	5

Project Title: SB Stocks - Paper Trading Platform

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Course: AI, ML and Gen AI Track

Project Overview

Abstract

SB Stocks is a web-based paper trading application that enables users to practice stock trading using virtual funds powered by real-time US stock market data. The platform provides a risk-free environment for beginners and experienced traders to test strategies, track portfolios, and build confidence.

Project Objectives

- 1. Develop a user-friendly paper trading platform
- 2. Integrate real-time stock market data

Scope

Web application covering user auth, stock data, paper trading, portfolio management, and analytics.

Milestone 1: Project Setup & Backend Foundation

Activity 1.1: Environment Setup

- Python 3.12, Flask framework, VS Code, Git/GitHub
- Virtual environment

Activity 1.2: Database Design

- User table: id, username, email, password_hash, balance, created_at
- Portfolio table: id, user_id, symbol, quantity, cost_basis, current_price

Activity 1.3: User Authentication

- Registration with form validation
- Password hashing (bcrypt)

Milestone 2: Stock Market Data Integration

Activity 2.1: Alpha Vantage API Setup

- API key configuration
- Rate limiting handling

Activity 2.2: Stock Search Implementation

- Endpoint: /api/stocks/search?q=<query>
- Returns matching stock symbols

Activity 2.3: Real-Time Quote Service

- Endpoint: /api/stocks/quote/<symbol>
- Returns: price, volume, market status

Activity 2.4: Historical Data

- Endpoint: /api/stocks/history/<symbol>
- Daily price data over a specified period

Milestone 3: Trading Engine Development

Activity 3.1: Buy Order Processing

- Validate user has sufficient virtual balance
- Fetch current market price

Activity 3.2: Sell Order Processing

- Validate user holds sufficient shares
- Fetch current market price

Activity 3.3: Transaction Logging

- Every trade creates a Transaction record
- Fields: symbol, price, quantity, timestamp

Milestone 4: Frontend Development

Activity 4.1: Base Template & Navigation

- Jinja2 template inheritance with base.html
- Responsive navigation bar

Activity 4.2: Dashboard Page

- Portfolio summary cards (total value, P&L, balance)
- Watchlist widget

Activity 4.3: Trading Interface

- Stock search with real-time suggestions
- Stock detail view

Activity 4.4: Portfolio Page

- Holdings table: symbol, shares, avg price, current price, P&L
- Color-coded gain/loss

Activity 4.5: Styling & Responsiveness

- Custom CSS with clean, professional aesthetics
- Bootstrap grid for layout

Milestone 5: Testing & Deployment

Activity 5.1: Unit Testing

- Auth module tests (registration, login, invalid credentials)
- Trading engine

Activity 5.2: Integration Testing

- End-to-end user flow: register -> login -> search -> buy -> portfolio -> sell
- API integration

Activity 5.3: Performance Testing

- Page load time benchmarks
- Concurrent use

Activity 5.4: Deployment

- Deployed on Render with Gunicorn WSGI server
- PostgreSQL da

Conclusion & Future Enhancements

Conclusion

SB Stocks successfully delivers a comprehensive paper trading experience with real-time market data, intuitive trading interface, and robust portfolio management. The platform empowers users to practice trading strategies without financial risk.

Future Enhancements

- 1. AI-powered stock recommendations using ML models
- 2. Advanced charting tools and technical indicators